# Marketing Research

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# Chapter 1

# Prerequisites

# Chapter 2

# Introduction

You can label chapter and section titles using `{#label}` after them, e.g., we can reference Chapter 2. If you do not manually label them, there will be automatic labels anyway, e.g., Chapter **??**.

Figures and tables with captions will be placed in `figure` and `table` environments, respectively.

```
par(mar = c(4, 4, .1, .1))
plot(pressure, type = 'b', pch = 19)
```

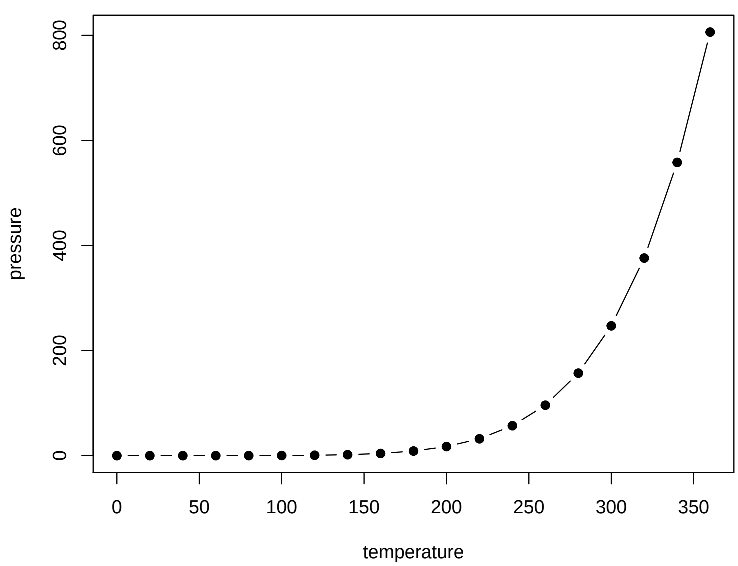



Figure 2.1: Here is a nice figure!

Table 2.1: Here is a nice table!

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|---:|---:|---:|---:|---|
| 5.1 | 3.5 | 1.4 | 0.2 | setosa |
| 4.9 | 3.0 | 1.4 | 0.2 | setosa |
| 4.7 | 3.2 | 1.3 | 0.2 | setosa |
| 4.6 | 3.1 | 1.5 | 0.2 | setosa |
| 5.0 | 3.6 | 1.4 | 0.2 | setosa |
| 5.4 | 3.9 | 1.7 | 0.4 | setosa |
| 4.6 | 3.4 | 1.4 | 0.3 | setosa |
| 5.0 | 3.4 | 1.5 | 0.2 | setosa |
| 4.4 | 2.9 | 1.4 | 0.2 | setosa |
| 4.9 | 3.1 | 1.5 | 0.1 | setosa |
| 5.4 | 3.7 | 1.5 | 0.2 | setosa |
| 4.8 | 3.4 | 1.6 | 0.2 | setosa |
| 4.8 | 3.0 | 1.4 | 0.1 | setosa |
| 4.3 | 3.0 | 1.1 | 0.1 | setosa |
| 5.8 | 4.0 | 1.2 | 0.2 | setosa |
| 5.7 | 4.4 | 1.5 | 0.4 | setosa |
| 5.4 | 3.9 | 1.3 | 0.4 | setosa |
| 5.1 | 3.5 | 1.4 | 0.3 | setosa |
| 5.7 | 3.8 | 1.7 | 0.3 | setosa |
| 5.1 | 3.8 | 1.5 | 0.3 | setosa |

Reference a figure by its code chunk label with the `fig:` prefix, e.g., see Figure 2.1. Similarly, you can reference tables generated from `knitr::kable()`, e.g., see Table 2.1.

```
knitr::kable(
  head(iris, 20), caption = 'Here is a nice table!',
  booktabs = TRUE
)
```

You can write citations, too. For example, we are using the **bookdown** package (Xie, 2020) in this sample book, which was built on top of R Markdown and **knitr** (Xie, 2015).

# Chapter 3

# Advertising

Prior distribution for TV advertising elasticity for consumer packaged goods can be found in (Shapiro et al., 2018). A substantial portion of products has statistically insignificant or negative estimates for advertising elasticity. TV ads might not be the best vehicle to reach the customer now.

# Chapter 4

# Brand Equity

Brand identity offers a value proposition to customers, where "a brand's value proposition is a statement of the functional, emotional, and self-expressive benefits delivered by the brand that provide value to the customers" (Aaker, 1996, pp.95). Functional benefits are product attributes that deliver functional utility to customers, while emotional benefits are positive feelings customers feel when using the brand. Ads that provide emotional benefits are higher in effectiveness score compared to functional benefits ads (Aaker, 1996, pp.98). For example, Evian – "Another day, another chance to feel healthy" ad - which provides emotional benefits of feeling satisfied after a workout. Moreover, self-expressive benefits are when brands provide a means for consumers to express their self-image. A person/ consumer can have multiple roles which associate with multiple self-concepts. Such as a woman, a mother, an author. The difference between emotional benefits and self-expressive benefits can be subtle. Emotional benefits focus on feelings, private setting (e.g., feeling of watching a TV show), past-oriented (memories), transitory, after-use emotional; In contrast, self-expressive benefits focus on self, public settings (e.g., using cars or clothes to signal), aspiration or future-oriented, permanent (i.e., the self links to person's personality), and during-use (wearing fancy clothes to signify a successful person). (cf. (Aaker, 1996, pp.99))

With the advancement in mass production of high -quality products, quality is no longer a dimension of status signal (Holt, 1998)

# Chapter 5

# Celebrity Endorsement

Based on Power Distance Theory, power distance beliefs moderates the positive effect of celebrity endorsers on evaluations of advertising (Winterich et al., 2018)

# Chapter 6

# WOM / Virality

81 percent of a telecom firm indicated that they are likely to refer other to the company; however, only 33 percent actually did. And of those who are referred, only 8 percent became profitable customer. (Kumar et al., 2007)

Even though suffering from the same fate as brand equity, WOM has made it come back under a different alias - virality. The attention that virality has garnered in recent years can be attributed to several reasons:

(1) the dawn of the Internet helps propel marketing into a new paradigm - digital marketing - where earned media is the new king

(2) the introduction of social network platforms such as Facebook, Twitter that facilitate the transmission of information and messages seamlessly and provide an unlimited source of data

(3) The advancement in network theory both theoretically, and empirically.

One platform that can be representative of this digital revolution is Twitter. According to a recent report, Twitter has 145 million monetizable daily active users. Twelve percent of Americans get their news from Twitter. And Twitter has 2 billion video impressions per day (25 Twitter Stats All Marketers Need to Know in 2020. There are more than 500 million Tweets are sent per day. (The 2014 #YearOnTwitter, 2014). One cannot deny the power of online WOM as well as its potential reach; hence, viral marketing has been born with these platforms.

The age group from 18-24 can be more challenging to target since they reported fewer positive responses compared to other age groups (Libert, 2014). Since their threshold is higher due to the overwhelming digital content they encounter, their emotions are harder to activate.

Wide gaps between two opposing opinions can increase future discussion (reviews); however, fluctuation around one opinion does not. Higher informational content moderate positively the relationship between opposing opinions and future reviews. (Nagle and Riedl, 2014)

WOM should be taken under the context of the evolution of conversation. (Berger, 2014)

**Valuable virality** (Akpinar and Berger, 2017)

Establishes causality of appeals and brand integralness to virality.

- Compared to informative appeal ads (focus on product features), emotional appeal ads (drama, mood, music, emotion-eliciting strategies) are more likely to be shared.

- Informative appeals boost brand evaluations and purchase because the brand is an integral part of the ad content.

- Hence, to combine the best of both worlds, emotional brand-integral ads boost sharing while also bolstering brand-related outcomes (embedded the brand into the stories).

- The mechanism that ads affect brand evaluation through two simultaneous mechanisms: persuasive attempt and brand knowledge.

"keeping viewers involved depends in large part on two emotions: joy and surprise"(Teixeira, 2012) - "Research: The Emotions that Make Marketing Campaigns Go Viral"

"ads that produce stable emotional states generally aren't effective at engaging viewers for very long." And "shock may get people to watch an ad privately, it often works against their desire to share the spot. Like Clothing Drive that try to mimic "Swear Jar" by Bud Light" (Teixeira, 2012)

(Westbrook, 1987, p. 261) defined WOM as "informational communication directed at other consumers about the ownership, usage, or characteristics of particular goods and services or their sellers." WOM evolves to "word of mouse" (or e-WOM).

(Berger, 2014):

WOM is goal-driven and serves five key functions:

- Impression Management

- Emotion regulation

- Information acquisition

- Social bonding
- Persuasion

The motivation behind this process is self -serving and even subconscious level. Hence, we can predict the types of news and information people are most likely to discuss.

- Over 70% of everyday speech is about the self, such as personal experiences or relationships (Dunbar et al., 1997). Consistent with previous research, online behaviors exhibit the same pattern: over 70% of social media posts are about self (Naaman et al., 2010).
- At any given point in time, multiple drivers can be present.

## 6.1 Impression Management

Consumers buy to signal their derived identities to achieve desired impressions (J. Berger & Heath, 2007)

Interpersonal communication (WOM) influences impression management in 3 ways: (1) self-enhancement, (2) identity -signaling, and (3) filling conversational space.

- **Self-enhancement**: human has a tendency to self-enhance (Markus, 1987). Hence, people like to share things to help them look good (Chung & Darke, 2006; Hennig-Thurau et al., 2004). Since bragging too much about oneself might have an opposite effect. People sometimes engage in" humblebragging": brag but self-deprecating at the same time. (Sezer et al., 2018)

- **Identity-signalling**: they talk about themselves to signal they have certain traits or expertise (G. Packard & Wooten, 2013), such as opinion leaders talk to signal their identity (share to show their knowledge).

- **Filling conversational space**: (small talk) people engage in small talk to avoid pauses and silence in between conversations so that the other party would not make lousy inferences about the person. (J. Berger, 2014)

Hence impression management induces people to share things that are

- **Entertaining** (i.e., interesting, surprising, funny or extreme) so that they shared look more interesting, and in-the-know. Interesting products get more word of mouth (J. Berger & Iyengar, 2013; J. Berger & Milkman, 2012; J. Berger & Schwartz, 2011). Moderate controversy is a catalyst for word of mouth to spread (Chen & Berger, 2013). And sometimes, people even exaggerate stories to make things more interesting (C. Heath, 1996); around 60% of the stories are distorted (Marsh & Tversky, 2004). Things that can be considered interesting are those that are either novel, exciting, or violate previous expectations (i.e., surprise) (Berlyne, 2006; Silvia, 2008). Nevertheless, comprehensibility can be a moderate of whether a novel thing is interesting: novel and comprehensible equals interesting, while novel and incomprehensible equal confusing (Silvia, 2008). Novelty refers to things that are new, surprising, exciting, or complex (Berlyne, 1960). Comprehensibility refers to the fact that the novelty must be understandable.

- **Useful**: because it makes the sharer smart and helpful. Empirical evidence show that useful stories(J. Berger & Milkman, 2012) and higher quality brands (Lovett et al., 2013) are more likely to be shared.

- **Self-concept relevant**: different people see different domain as the optimal signal amplifier such as politics, economics, etc. hence symbolic products are more likely to be shared than utilitarian ones (Chung & Darke, 2006) (even though useful information is helpful, but the signal to be smart and helpful are not appreciated as being interesting and entertaining).

- **Status related**: premium brands are talked about more (Lovett et al., 2013). Knowledge is can also be considered as a status symbol, and people share to signal that they re in the know (Ritson & Elliott, 1999)

- **Unique**: unique products are more likely to be shared, but people with a high need for uniqueness will be less likely to share positive WOM because they do not want people to adopt the product and reduces their uniqueness (Cheema & Kaikati, 2010). Sometimes they even scare others by citing product complexity (Moldovan et al., 2015)

- **Common ground**: sharing common things induces listener to convey that there is interpersonal similarity and facilitates conversation.

- **Emotional valence**: people prefer to share positive things than negative news (J. Berger & Milkman, 2012). However, in some special cases, negative WOM can be perceived as a desired component such as reviewers are seen as more intelligent, (Amabile, 1983). Which is moderated by whether people are refereeing to themselves or others will affect WOM valence (positive vs. negative)(Kamins et al., 1997). Positive WOM when talking oneself conveys a positive image, while negative WOM about others convey that they are relatively better than the other party.

- **Incidental arousal**: unrelated arousal (e.g., running in place) can increase sharing in general (J. Berger, 2011) because of the increase in arousal levels.

- **Accessible things**: products with more cued or triggered will be discussed more (J. Berger & Schwartz, 2011). Hence, more advertised products generate more WOM (Onishi & Manchanda, 2012). "top of mind" and "tip of the tongue." Availability bias means that the easier we can recall something, the more likely we are going to talk bout it. Publicly visible products also have more WOM (J. Berger & Schwartz, 2011)

## 6.2 Emotion regulation

WOM help consumers regulate their emotions. Emotion regulation refers to "a person's ability to effectively manage and respond to an emotional experience" (Rolston & Lloyd-Richardson, 2017). and sharing with other (WOM) can facilitate emotion regulation by:

- **Generating social support**: sharing can give you comfort and consolation (Rimé, 2009). For example, after a negative emotional experience, sharing can help increase well-being through perceived social support (J. A. Berger & Buechel, 2012)

- **Venting**: WOM (sharing) helps people achieve catharsis when experiencing negative experiences(Alicke et al., 1992): consumers express when they are angry (Wetzer et al., 2007) or dissatisfied (Anderson, 1998).

- **Sensemaking**: "talking can help people understand what they feel and why" (J. Berger, 2014)

- **Reducing dissonance**: even after making a decision, we want to make sure we make the right choice by talking to others tot reduce feelings of doubt (Rosnow, 1980)

- **Taking vengeance:** to punish the company (different from venting, which makes you feel better), consumers share negative consumption experience (Ward & Ostrom, 2006)

- **Rehearsal**: people talk to relive positive experiences (Rimé, 2009)

Emotion regulation drives people to share (1) emotional content (2) influence the valence of the content shared, (3) lead people to share more emotionally arousing content:

- **Emotionality**: more emotional intensity, more sharing (J. Berger & Milkman, 2012), people share more emotional social anecdotes (Peters et al., 2009). But not all emotions increase sharing: for example, shame and guilt decreases sharing (Finkenauer & Rimé, 1998) since sharing those things makes people look bad.

- **Valence**: "Emotion regulation tends to focus on the management of negative emotions" (J. Berger, 2014), but a counter vailing effect is that under impression management, people avoid sharing negative stories since it might reflect on the shares. Sharing negative things can decrease willingness to share (Chen & Berger, 2013)

- **Emotional arousal**:
  - **Negative** side: low arousal emotion (sadness), high arousal (anger or anxiety) should increase the need to vent.

  - **Positive** side: low arousal (contentment), high arousal (awe, excitement or amusement) increase desires for rehearsal.

## 6.3 Information acquisition

WOM helps acquire information.

- **Seeking advice**: (Hennig-Thurau et al., 2004; Rimé, 2009). Rather than trial and error, or direct observation, gossip serves as another form of learning (Baumeister et al., 2004)

- **Resolving problems**: online forums or other online opinion platforms help customers find a solution quickly from others (Hennig-Thurau et al., 2004)

Hence the process of information acquisition would induce to talk more about

- **Risky, important, complex, or uncertainty-ridden decision**: talking to others can reduce risk, simplify complexity, and increase consumer's confidence (Hennig-Thurau et al., 2004)

- **Lack of (trustworthy) information**:

## 6.4 Social bonding

"people have a fundamental desire for social relationships (Baumeister & Leary, 1995), and interpersonal communication fills that need (Hennig-Thurau et al., 2004)". Empirically, people in brand communities because want to connect with others like them (Muniz & O'Guinn, 2001). Sharing facilitates social bonding through

- **Reinforce shared views**: group memberships, and one's place in the social hierarchy. WOM (sharing) allows people to fit in as a group member (J. Berger & Heath, 2007; Ritson & Elliott, 1999)

- **Reducing loneliness and social exclusion**: sharing decrease interpersonal distance and decreases loneliness and exclusion (Maner et al., 2007)

Social bonding drives what people share: people share things are :

- **Common ground**: social bonding drives people to talk about things they have in common because it makes them feel socially connected (Clark & Kashima, 2007)
- **Emotionality**: "sharing an emotional story or narrative increases the chance that others will feel similarly" (J. Berger, 2014). Emotional similarity increase group cohesiveness (Barsade & Gibson, 2007). And emotion sharing and social bonding can be a simultaneous process since previous works found that emotion sharing bonds people (Peters & Kashima, 2007) and high arousal emotion increase social bonding needs (Chen & Berger, 2013)

## 6.5 Persuading others

people use WOM to persuade others to affect their satisfaction or choices (J. Berger, 2014). Persuasive drive people to share things that are

- **Emotionally polarized** (polarized valence): since the goal is to convince something is good or bad (i.e., people share extremely rather than moderately positive(negative) information

- **Arousing content**: "people who want to persuade others may find shared arousing content to incite others to take desired actions" (J. Berger, 2014)

## 6.6 Moderators

- Two key moderators when different WOM functions have a greater impact:
  - **Communication audience**:
    - Tie strength
    - Audience size
    - Tie status
  - **Communication Channel**
    - Written vs. oral
    - Identifiability: whether communicators are identifiable. (anonymously)

    - Audience salience: whether the audience is salient during communication (online communication, the audience is less salient).

Social motivation: why we share :(why_some_videos_go_viral_2015)

- Impression management:
  - authority (e.g., “I want to demonstrate my knowledge”)
  - Social utility (e.g., this could be useful to my friends)
  - coolhunter (e.g., I want to be the first o tell my friends)
  - Zeitgeist (e.g., It’s about a current trend or event)
  - Conversation starting (e.g., I want to start an online conversation)
  - Self-expression (e.g., it says something about me)
  - Social good (e.g., It’s for a good cause, and I want to help”).
- Information acquisition:
  - Opinion seeking (e.g., “I want to see what my friends think)
- Social bonding:
  - Shared passion (e.g, “it lets me connect with my friends bout a shared interest)
  - Social in real life (e.g., it will help me socialize with my friends offline).

More frequent exposure to perceptually or conceptually related cues increases product accessibility and makes the product easier to process. In turn, this increased accessibility influences product evaluation and choice, which are found to vary directly with the frequency of exposure to conceptually related cues. These results support the hypothesis that conceptual priming effects can have a strong impact on real-world consumer judgment” (Berger and Fitzsimons, 2008)

# Chapter 7

# CLV

“A year’s projected business gives a number that is normally about half of a customer’s full lifetime value, although that can vary by industry.” (Kumar et al., 2007)

(Kumar et al., 2008) CLV model

$$CLV_i = \sum_{j=T+1}^{T+N} \frac{p(Buy_{ij} = 1) \times \hat{CM}_{ij}}{(1+r)^{j-T}} - \frac{\hat{MC}_{ij}}{(1-r)^{j-T}}$$

where

- $CLV_i$ = lifetime value for customer i,
- $p(Buy_{ij})$ = predicted probability that customer i will purchase in period j,
- $CM_{ij}$ = predicted contribution margin provided by customer i in period j,
- $MC_{ij}$ = predicted marketing costs directed toward customer i in period j,
- $j$ = index for time periods (semiannual or annual)
- $T$ = the end of the calibration or observation time frame (usually 1 year projection),
- $N$ = total number of prediction periods,
- $r$ = semiannual discount factor (e.g., 0.07238 (Kumar et al., 2008) = 15% annual rate)

## 7.1 Referral value (CRV)

"Referrals made by customers after a referral-incentive marketing campaign can be attributed to that campaign for about a year." (Kumar et al., 2007). Hence, we usually count those referrals made after one year of a campaign.

2 types of referral:

- Type-one referral: Would not join without referral.
- Type-two referral: still join without referral.

(Kumar et al., 2007) estimated that each customer made about half type-one and half type-two referrals.

Referral value = PV(Type-one referrals) + PV(Type-two referrals)

where

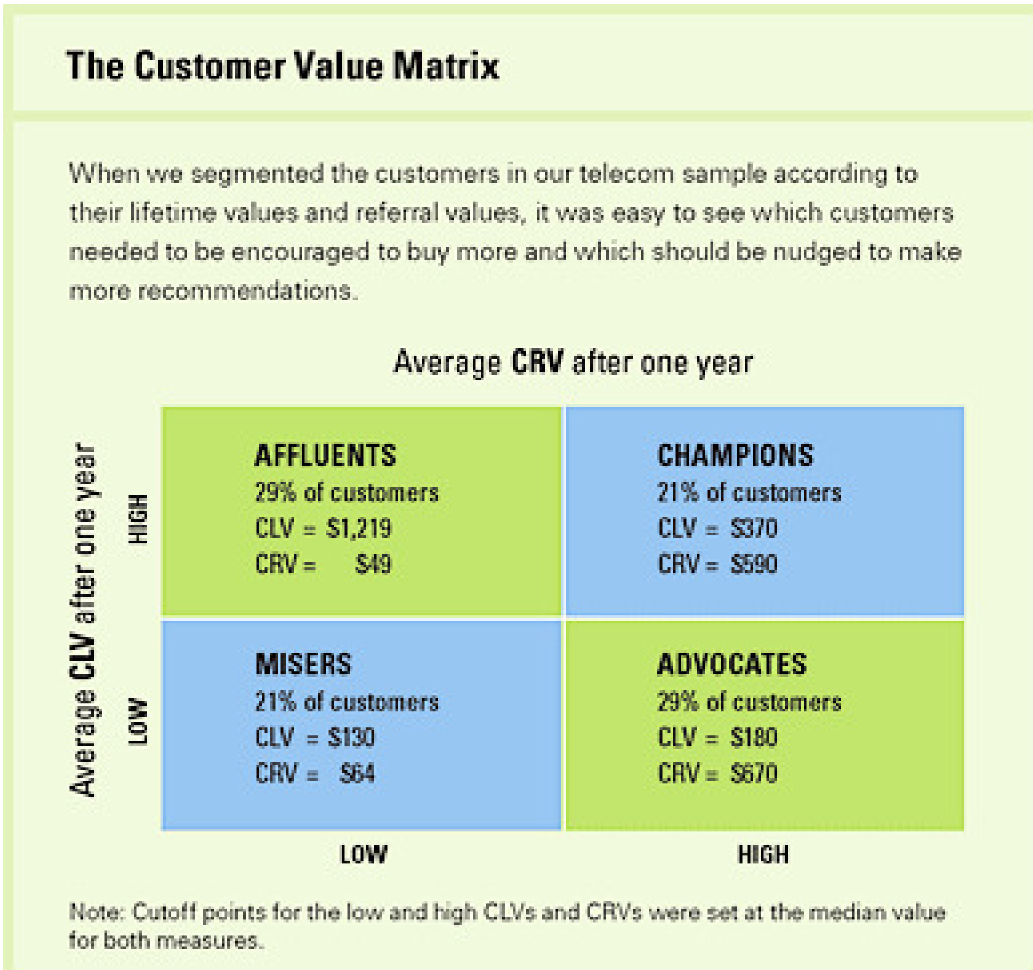
- PV(type-two referrals) = the prevent value of the savings in acquisition costs.

$$CRV_i = \sum_{t=1}^{T} \sum_{y=1}^{n1} \frac{A_{ty} - a_{ty} - M_{ty} + ACQ1_{ty}}{(1+r)^t} + \sum_{t=1}^{T} \sum_{y=n1+1}^{n2} \frac{ACQ2_{ty}}{(1+r)^t}$$

where

- T = the number of periods that will be predicted into the future (e.g., typically 1 year)
- $A_{ty}$ = the gross margin contributed by customer y who other otherwise would not have bought the product,
- $a_{ty}$ = the cost of the referral for the customer y (this is what you set up)
- $n1$ = the number of customers who would not join without the referral,
- $n2 - n1$ = the number of customers who would have joined anyway,
- $M_{ty}$ = the marketing costs needed to retain the referred customers,
- r = semiannual discount factor ((Kumar et al., 2008) used .07238 = 15% annual rate)
- $ACQ1_{ty}$ = the savings in acquisition costs from customers who would not join without the referral,
- $ACQ2_{ty}$ = the savings in acquisition cost from customers who would have joined anyway.

Customers with high CLV does not guarantee high CRV (Kumar et al., 2007). Hence, (Kumar et al., 2007) proposed the customer value matrix



# Chapter 8

# Finance-Marketing Interface

## 8.1 Tobin's q

Retrieve data from WRDS

```
# to set up connection from R to WRDS (https://wrds-www.wharton.upenn.edu/pages/support/programmi
library(RPostgres)
library(dplyr)

# I've set up wrds connection before hand. Please use your username and password here.

# wrds <- dbConnect(Postgres(),
#                   host='wrds-pgdata.wharton.upenn.edu',
#                   port=9737,
#                   dbname='wrds',
#                   sslmode='require',
#                   user='')

# Check variables (column headers) in COMP ANNUAL FUNDAMENTAL
#uses the already-established wrds connection to prepare the SQL query string and save the query
# check avaiable databases: https://wrds-web.wharton.upenn.edu/wrds/tools/variable.cfm?vendor_id=
res <- dbSendQuery(wrds, "select column_name
                   from information_schema.columns
                   where table_schema='compa'
                   and table_name='funda'
                   order by column_name")
data <- dbFetch(res, n=-1) # fetches the data that results from running the query res against wrd
dbClearResult(res) # closes the connection
```

```
head(data)
```

```
##   column_name
## 1       acchg
## 2        acco
## 3       accrt
## 4     acctchg
## 5     acctstd
## 6        acdo
```

```
# select everything
res <- dbSendQuery(wrds, "select * from compa.funda")

# from compa.funda

# only select the following variables
res <- dbSendQuery(wrds, "select gvkey, datadate, fyear, indfmt, consol, popsrc, datafr
```

```
## Warning in result_create(conn@ptr, statement): Closing open result set,
## cancelling previous query
```

```
data1 <- dbFetch(res, n=-1)
dbClearResult(res)

data = data1 %>%
    distinct(gvkey,datadate,fyear,tic,conm,.keep_all = T)
```

Calculate Tobin's Q

```
tobin_q_dt = data %>%
    mutate(mkvalt = coalesce(mkvalt,0),
           dltt = coalesce(dltt,0),
           at = coalesce(at,0),
           dlc = coalesce(dlc,0),
           act = coalesce(act,0)) %>%
    mutate(tobin_q = (mkvalt + ifelse((dlc - act) >=0,as.numeric(dlc-act),0) + dltt)/at
    select(tobin_q,gvkey,datadate,fyear,conm)
head(tobin_q_dt)
```

```
##     tobin_q  gvkey   datadate fyear                  conm
## 1       Inf 001000 1961-12-31  1961 A & E PLASTIK PAK INC
## 2       NaN 001000 1962-12-31  1962 A & E PLASTIK PAK INC
## 3       Inf 001000 1963-12-31  1963 A & E PLASTIK PAK INC
## 4 0.3686441 001000 1964-12-31  1964 A & E PLASTIK PAK INC
## 5 0.4995671 001000 1965-12-31  1965 A & E PLASTIK PAK INC
## 6 0.4563786 001000 1966-12-31  1966 A & E PLASTIK PAK INC
```

## 8.2 Event study

### 8.2.1 Abnormal Return

https://rpubs.com/gregoryarobinson/315096

https://www.eventstudytools.com/introduction-event-study-methodology

```
library(EventStudy)
```

Another package

https://cran.r-project.org/web/packages/eventstudies/vignettes/replications.pdf

```
library(eventstudies)
```

Replication of (Chen and Siems, 2004)

```
data(TerrorIndiceReturns)
data(TerrorAttack)
TerrorIndiceCAR <- lapply(1: ncol(TerrorIndiceReturns), function(x){
# 10-day window around the event
event <- phys2eventtime(na.omit(TerrorIndiceReturns[ , x,
drop = FALSE]),
TerrorAttack,
10)
# Estimate ARs
esMean <- constantMeanReturn(event$z.e[which(attributes(event$z.e)$index
%in% -30:-11), ],
residual = FALSE)
ar <- event$z.e - esMean
ar <- window(ar, start = 0, end = 10)
# CAR
car <- remap.cumsum(ar, base = as.numeric(ar[1, 1]))
names(car) <- colnames(TerrorIndiceReturns[ , x,
drop = FALSE])
return(car)
})
names(TerrorIndiceCAR) <- colnames(TerrorIndiceReturns)
# Compile for all indices
TerrorIndiceCAR <- do.call(cbind, TerrorIndiceCAR)
# 11-day CAR
TerrorIndiceCAR[11, ]
```

```
##       S.P500       Dow      NYSE    NASDAQ   Toronto    Mexico    London
## 10 -3.832794 -7.897366 -3.981837 -9.993519 -9.110412 -6.53973 -8.602633
##      Germany Eurostoxx50    France     Spain Switzerland   Austria     Italy
## 10 -10.45718   -7.596024 -9.518721 -10.6724   -7.292201 -7.894399 -15.21462
```

```
##      Belgium Amsterdam Portugal  Helsinki    Norway    Sweden     Tokyo
## 10 -9.219016 -10.82756 1.698079 -5.500048 -12.39281 -4.691903 -3.603373
##     HongKong   S.Korea     India   Jakarta Singapore Kualalampur Australia
## 10 -5.231521 -16.64571 -17.02777 -9.933654 -17.98219   -14.55417 -8.597988
##          NZL  Pakistan     Saudi   Israel     Jberg
## 10 -5.635766 -12.42189 -11.32466 -7.81159 -12.59022
```

```
TerrorIndiceCAR[6, ] # 6-day CAR
```

```
##     S.P500       Dow      NYSE    NASDAQ  Toronto    Mexico    London   Germany
## 5 -7.72379 -10.56695 -8.094739 -10.13781 -7.01292 -13.17444 -4.533527 -8.063819
##   Eurostoxx50    France     Spain Switzerland   Austria     Italy   Belgium
## 5   -5.940704 -8.707865 -9.426443   -5.967055 -4.428614 -13.98185 -8.510382
##   Amsterdam  Portugal  Helsinki    Norway   Sweden      Tokyo  HongKong
## 5 -8.521101 -6.875878 -4.802975 -9.894391 -4.96208 -0.5636291 -5.570385
##     S.Korea     India  Jakarta Singapore Kualalampur Australia       NZL
## 5 -11.81901 -11.77841 -4.91951 -13.74157   -10.50214 -6.814062 -7.101218
##   Pakistan     Saudi    Israel     Jberg
## 5 -10.3741 -5.320904 -11.03399 -12.09548
```

# Chapter 9

# Advertising

People don't like intrusive ads, especially when they are vulnerable (Tybout and calkins, 2005, p.143)

# Chapter 10

# Data

```
knitr::opts_chunk$set(eval = FALSE)
```

## 10.1 WRDS

Retrieve data from WRDS

```
# to set up connection from R to WRDS (https://wrds-www.wharton.upenn.edu/pages/support/programm
library(RPostgres)
library(dplyr)

# I've set up wrds connection before hand. Please use your username and password here.

# wrds <- dbConnect(Postgres(),
#                   host='wrds-pgdata.wharton.upenn.edu',
#                   port=9737,
#                   dbname='wrds',
#                   sslmode='require',
#                   user='')

# Check variables (column headers) in COMP ANNUAL FUNDAMENTAL
#uses the already-established wrds connection to prepare the SQL query string and save the query
# check avaiable databases: https://wrds-web.wharton.upenn.edu/wrds/tools/variable.cfm?vendor_id=
res <- dbSendQuery(wrds, "select column_name
                   from information_schema.columns
                   where table_schema='compa'
                   and table_name='funda'
                   order by column_name")
data <- dbFetch(res, n=-1) # fetches the data that results from running the query res against wr
```

```
dbClearResult(res) # closes the connection
head(data)

# select everything
res <- dbSendQuery(wrds, "select * from compa.funda")

# from compa.funda

# only select the following variables
res <- dbSendQuery(wrds, "select gvkey, datadate, fyear, indfmt, consol, popsrc, datafr
data1 <- dbFetch(res, n=-1)
dbClearResult(res)

data = data1 %>%
    distinct(gvkey,datadate,fyear,tic,conm,.keep_all = T)
```

## 10.2 YouTube

### 10.2.1 OAuth

Go to link to access your API

```
library(tuber)
app_id = "YOUR APP ID"
app_secret = "YOUR APP SECRET"
yt_oauth(app_id, app_secret)


get_stats(video_id = "N708P-A45D0")
```

Note for Ubuntu user

```
httr::set_config(httr::config( ssl_verifypeer = 0L ) )
```

Get info about a video

```
get_video_details(video_id="N708P-A45D0")
```

Get caption of a video

```
get_captions(video_id="yJXTXN4xrI8")
```

Search Videos

```
res <- yt_search(term = "test")
head(res[, 1:3])
```

Get Comments of a video

```
res <- get_comment_threads(c(video_id="N708P-A45D0"))
head(res)
```

Get All the Comments Including Replies

```
get_all_comments(video_id = "a-UQz7fqR3w")
```

Get statistics of all the videos in a channel

```
a <- list_channel_resources(filter = c(channel_id = "UCT5Cx1l4IS3wHkJXNyuj4TA"), part="contentDet

# Uploaded playlists:
playlist_id <- a$items[[1]]$contentDetails$relatedPlaylists$uploads

# Get videos on the playlist
vids <- get_playlist_items(filter= c(playlist_id=playlist_id))

# Video ids
vid_ids <- as.vector(vids$contentDetails.videoId)

# Function to scrape stats for all vids
get_all_stats <- function(id) {
  get_stats(id)
}

# Get stats and convert results to data frame
res <- lapply(vid_ids, get_all_stats)
res_df <- do.call(rbind, lapply(res, data.frame))

head(res_df)
```

### 10.2.2 API

```
require(curl)
require(jsonlite)
library(kableExtra)
library(dplyr)
library(ggplot2)
library(plotly)
library(reshape2)

API_key = "YOUR API KEY"

getstats_video<-function(video_id,API_key){

  url=paste0("https://www.googleapis.com/youtube/v3/videos?part=snippet,statistics&id=",video_id,
```

```
  result <- fromJSON(txt=url)
  salida=list()
  return(data.frame(name=result$items$snippet$channelTitle, result$items$statistics,ti
}

get_playlist_canal<-function(id,API_key,topn=15){

  url=paste0('https://www.googleapis.com/youtube/v3/playlistItems?part=contentDetails&p

  result=fromJSON(txt=url)
  return(data.frame(result$items$contentDetails))
}

getstats_canal<-function(id,API_key){

url=paste0('https://www.googleapis.com/youtube/v3/channels?part=snippet%2CcontentDetai

result <- fromJSON(txt=url)
return(data.frame(name=result$items$snippet$title,result$items$statistics,pl_list_id=re

}

getall_channels<-function(ids,API_key,topn=5){

  videos=lapply(ids,FUN=get_playlist_canal,API_key=API_key,topn=topn) %>% bind_rows()

  stats=lapply(videos[,1],FUN=getstats_video,API_key=API_key)
  stats=bind_rows(stats)
  stats$vid_id=videos[,1]
  return(stats)
}
```

Statistics per Channel

```
can_st=lapply(comp_data$cha_id,FUN = getstats_canal,API_key=API_key)
can_st=bind_rows(can_st)
can_st$viewCount=as.numeric(can_st$viewCount)
can_st[,1:6] %>% kable() %>%kable_styling()
```

```
can_st$viewCount=round(as.numeric(can_st$viewCount)/1000000,2)
p1=can_st %>% ggplot(aes(x=reorder(name,viewCount),y=viewCount,fill=name))+
geom_bar(stat="sum")+guides(size=F)+coord_flip()+scale_fill_manual(values = c("red", "c
geom_text(inherit.aes = T,aes(label=paste(viewCount,"M")),nudge_y =0,angle = 90)+
labs(x="Total Visualizations(Millions)",y="Visualizations",fill="")+
theme(legend.position = "top")+mytheme3
```

```
  ggplotly(p1,tooltip=c("name","viewCount"))  %>%
    layout(legend = list(orientation = "h",x = 0.01, y = -0.1,autosize=F))
```

Information on Individual Videos per Channel

```
var_to_see="dislikeCount" #favoriteCount or commentCount

info=getall_channels(ids = can_st$pl_list_id.uploads,API_key = API_key,topn =20)


 datacond=melt(info[,c(1:6,8)],id.vars = c("name","date"))
 datacond$date=as.Date(datacond$date)
 datacond$value=as.numeric(datacond$value)


ggplot(filter(datacond,variable==var_to_see),aes(x=as.Date(date),y=value,fill=name))+
geom_bar(stat="sum")+labs(x=var_to_see,y="",fill="")+guides(size=FALSE)+scale_fill_manual(values
  scale_x_date(limits =as.Date(c(as.Date(min(datacond$date)),as.Date(Sys.time()))),date_breaks  =
```

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